Team EV — CptS 484 Project THEIA

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1. About THEIA

THEIA is a smartphone application designed to help blind and visually impaired users navigate indoor campus buildings. The application uses voice commands, spoken directions, and vibrations to guide users to their destinations and provide assistance when they become lost.

THEIA works alongside your cane or guide dog rather than replacing them.

What you need:

- A smartphone with Bluetooth enabled.
- A campus building supported by THEIA.
- Earphones are recommended so you can hear instructions clearly.

How to control THEIA:

THEIA can be controlled using voice commands or touchscreen gestures. You do not need to see the phone screen to use the application. THEIA will respond with both speed and vibration allowing the users to receive a message without needing to see the screen.

To begin, open THEIA using your phone's accessibility features or voice assistant.

2. First-Time Setup (Caretaker)

A caretaker should help configure THEIA before the user begins navigating.

1. Open THEIA and go to **Screen 5: Settings**.
2. Select the caretaker settings and enter or create a PIN.
3. Add emergency contacts, including their names and phone numbers.
4. Choose whether to enable automatic fall detection.
5. Select a default navigation mode: Fast, Safe, or Comfortable.
6. Enable the "No stairs" preference if needed.
7. If step wanted, enable and finish step calibration of the user to help better predict distance.
8. Optionally, add frequently visited locations or the user's class schedule.

The user can change comfort settings such as volume, speech speed, and vibration strength. Caretaker safety settings are protected by a PIN.

3. Going to a Room (Scenario 1: Stevie)

1. Open THEIA. The Voice Home screen asks, "Where would you like to go?"
2. Tap anywhere on the screen and say your destination, such as "Take me to Spark 233."
3. THEIA repeats your destination and asks, "Spark 233. Is that right?"
4. Say "Yes" or double-tap to confirm. You will feel one short vibration.
5. Follow THEIA's spoken directions, such as "Walk 20 steps, then turn left."
6. Approximately 5 steps before turning, THEIA says, "In 5 steps, turn left."

7. When you reach the turn, THEIA says, "Stop. Turn left now." You will feel one short vibration.
8. Continue following directions until THEIA announces, "Spark 233 is on your right."
9. One long vibration indicates that you have reached your destination.

Tips:

- Say "Repeat" or double-tap to hear the previous instruction again.
- Say "Where am I?" or use the location gesture to hear your current building, floor, and nearest room.

4. Changing Floors by Elevator (Scenario 3: Mark)

1. Open THEIA and say, "Take me to TFLO 131 via elevator."
2. THEIA identifies available routes and announces up to three options.
3. Say the number of your preferred route.
4. Follow the spoken directions until you reach the elevator.
5. THEIA announces the elevator location and provides instructions such as "Call button on your right."
6. Press the elevator call button and enter when it is safe.
7. THEIA says, "Go to floor 1."
8. Select the correct floor using the elevator controls.
9. After leaving the elevator, allow THEIA to identify your location and continue providing directions to prevent getting off on the wrong floor.
10. Follow the remaining instructions until you reach your destination.

If you miss an instruction, double-tap to hear it again.

5. If You Are Lost or Need Help (Scenario 2: Alfred)

This scenario demonstrates how THEIA assists you when you leave your planned route, encounter a blocked hallway, or need help from a caretaker.

5.1 Your Way Is Blocked

1. If you encounter construction or another blocked pathway, stop walking.
2. Say "This way is blocked."
3. THEIA responds, "OK. Finding a detour."
4. Wait while THEIA attempts to identify another route.
5. If a new route is available, listen to the directions and continue navigating.
6. If THEIA cannot identify a safe route, remain in a safe location and request assistance.

5.2 THEIA Says "You May Be Off Route"

1. If you miss an expected navigation checkpoint by approximately 30 seconds longer than expected, THEIA triggers the lost flag.
2. You will feel three vibrations and hear "You may be off route."
3. THEIA says, "Say 1 to find a new route, or 2 to go back to Spark 233."
4. Choose one of the following:
    - Say "1" or tap the upper half of the screen to request a new route.
    - Say "2" or tap the lower half of the screen to return to your last known location.
5. Follow THEIA's spoken recovery instructions.
6. If THEIA cannot identify your current location, stop in a safe place rather than continuing without reliable directions.

7. If you remain lost for 2 minutes, THEIA asks whether you want to send a message to your caretaker.
8. Say "Yes" to request assistance or "No" to decline.

THEIA does not automatically contact your caretaker after 2 minutes. It first asks for your permission.

5.3 Asking for Help

1. Say "Help" or hold two fingers on the screen for 2 seconds.
2. THEIA opens **Screen 4: Help** and provides a strong vibration.
3. THEIA asks, "Do you need help? Text caretaker, call caretaker, call campus police, or cancel?"
4. Choose one of the following options:
    - **Text caretaker:** Sends an assistance request.
    - **Call caretaker:** Calls your registered caretaker.
    - **Call emergency contact:** Initiates a call for campus assistance or other emergency contact.
    - **Cancel:** Returns to the previous activity.
5. If you select "Text caretaker," THEIA sends available location details such as your building, floor, nearest known room, time, and a map link.
6. Your caretaker can use this information to help locate you.
7. If your caretaker responds, a vibration along with a voice memo of the message along with a bubble of THEIA showing the message reply on screen
8. If your caretaker does not respond, THEIA offers the option to contact campus police.

Example:

Alfred is navigating to his destination when he encounters construction blocking the hallway. He takes a detour and becomes lost. THEIA detects that he has missed a checkpoint and offers to reroute him or return him to Spark 233.

When Alfred cannot recover his route, he requests help. THEIA provides the option to text his caretaker, Bella, with his last known location so she can assist him.

5.4 If You Fall

1. If THEIA detects a possible fall, it opens the Help fall-check screen.
2. THEIA provides a strong continuous vibration.
3. THEIA asks, "Are you OK? I will text Bella in 30 seconds. Say Cancel to stop."
4. A 30-second countdown begins.
5. If you are okay, say "Cancel" to stop the countdown.
6. If you do not respond within 30 seconds, THEIA automatically sends a text message to Bella.
7. THEIA then attempts to contact your first emergency contact.

This is a proposed prototype feature and should not be treated as a guaranteed emergency service.

6. Changing Settings and Modes (Scenario 4: Elena)

6.1 Changing User Settings

THEIA allows you to adjust settings using voice commands.

1. Open THEIA.
2. Say a command such as "THEIA, louder."
3. THEIA responds, "Volume 80 percent."

4. You can also change speech speed, how often instructions are given, and vibration strength.
5. THEIA confirms changes through spoken feedback.

6.2 Navigation Modes

THEIA provides three navigation modes.

Mode	Description
Fast	Prioritizes shorter or faster routes.(congestion)
Safe	Prioritizes routes that avoid known hazards. (congestion)
Comfortable	Prioritizes routes based on user preferences. (ignores congestion)

To change modes, say "Fast mode," "Safe mode," or "Comfortable mode."

Caretaker Settings

Emergency contacts, fall detection, default navigation mode, and the "No stairs" preference are managed through PIN-protected caretaker settings.

7. Quick Reference Card

Voice Commands

“User says ____”	Theia responds with this
"Take me to ___"	Starts a route
"Where am I?"	Tells building, floor, nearest room, which way you face
"Repeat"	Says the last instruction again
"This way is blocked"	Finds a detour
"Help"	Opens the Help screen
"Cancel"	Stops the current action or countdown
“Volume down/up, speed slower/faster”	Changes volume of voice and speed of speech
"Fast mode" / "Safe mode" / "Comfortable mode"	Changes route style to pre planned or programmed routes based on info

Touchscreen Gestures

“User says ____”	Theia responds with this
"Take me to ___"	Starts a route
"Where am I?"	Tells building, floor, nearest room, which way you face
"Repeat"	Says the last instruction again
"This way is blocked"	Finds a detour
"Help"	Opens the Help screen
"Cancel"	Stops the current action or countdown
“Volume down/up, speed slower/faster”	Changes volume of voice and speed of speech
"Fast mode" / "Safe mode" / "Comfortable mode"	Changes route style to pre planned or programmed routes based on info

Vibration Alerts

“User says ____”	Theia responds with this
"Take me to ___"	Starts a route
"Where am I?"	Tells building, floor, nearest room, which way you face
"Repeat"	Says the last instruction again
"This way is blocked"	Finds a detour
"Help"	Opens the Help screen
"Cancel"	Stops the current action or countdown
“Volume down/up, speed slower/faster”	Changes volume of voice and speed of speech

"Fast mode" / "Safe mode" / "Comfortable mode"	Changes route style to pre planned or programmed routes based on info

8. Limitations of THEIA

- THEIA is designed for indoor navigation within supported campus buildings.
- THEIA does not provide navigation to objects inside individual rooms, such as chairs or desks.
- THEIA does not detect moving pedestrians or vehicles.
- Continue using your cane or guide dog when navigating.
- Emergency calls and text messages require an available phone connection.
- Navigation accuracy depends on available building information and location checkpoints.
- THEIA is an assistive navigation application and does not replace normal mobility and safety precautions.

End of Preliminary User Manual